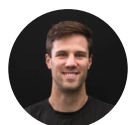


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August 4, 2025

Bar path tracking | AR video playback for technique review

Metric VBT – Free bar path tracker. Record and save video with bar path, velocity, ROM and tempo data analytics.



Jacob Tober
Co-founder



The Metric VBT app now includes bar path tracking

The bar path feature is available for free in the Metric iOS app (sorry

Android users, coming soon!).

[Download Metric on your iPhone here →](#)

VBT Coach - Velocity Based Training Made Practical





Bar path tracking is an often misunderstood and underrated tool in strength and power training. Smart lifters and coaches leverage bar path tracking to accelerate technical improvement, ultimately leading to more PRs than ever.

What does bar path mean?

Bar path is a visual tool used in the gym to refine and analyse lifting technique. It is the trajectory or line that a barbell takes during a repetition.

To capture your bar path, you need some form of barbell tracking app like Metric VBT. Using Metric, you record a video of your lift from roughly side on, and can then playback or export the video with an overlay of your bar path to see a visualisation of how the barbell is moving in space.

What is the correct bar path?

The path a bar takes will vary depending on the type of lift being performed.

Generally speaking, however, there is not a "perfect" path for all people. There are undoubtedly bad paths that your bar can take during a rep, but the ideal path your barbell will trace will be unique and may even evolve over time as you get stronger and improve your technique.

What is the most efficient bar path?

Instead of thinking about mimicking an "ideal," good coaches and lifters focus on developing a consistent path that maximises their performance.

The most effective and efficient bar path also varies a lot by exercises, anatomy and training goals.

Bench press bar path

The path your barbell will follow on a bench press is typically more of an arc rather than a straight line, and the eccentric and concentric paths usually vary.

After unracking, the bar should start straight over your shoulders, descend in an arc towards your chest, and then press back up in a slight arc towards your shoulders. This arc allows for better engagement of the chest muscles and helps to protect your shoulders from unnecessary strain.

Bar path
on Bench
Press –
Starting
point and
end point
are the
same but
the
eccentric
and
concentric
paths
taken are
very
different

Things can get really technical on [bar path for the bench press](#), this article is another deep dive I found really interesting.

Squat bar path

For the squat, the bar path should be as close to a straight line as possible. The bar should remain over your midfoot throughout the entire movement. As you descend, focus on keeping your chest up and your back straight. Drive through your heels as you stand back up, ensuring the bar stays in line with your midfoot.

A bar path that drift towards the front or back of the foot during your repetitions can be a key clue to potential weaknesses or technical bias you might have in squatting.

Deadlift bar path

Similar to the squat, the deadlift bar path should be a straight line. The bar should start over your midfoot, and as you lift, it should remain close to your body, almost grazing your shins and thighs. Keep your back straight and engage your core throughout the movement to maintain proper form.

Bar path tracking for Olympic Weightlifting

For explosive lifts such as cleans, jerks, and snatches, bar path tracking becomes a mission-critical training tool. Olympic weightlifting movements require extreme precision, timing, and coordination to successfully complete a lift. Any deviation in the bar path can have huge repercussions and be the difference between a successful lift and a failed attempt.

Bar path tracking for weightlifting allows advanced weightlifters and their coaches to dissect every section of their lift, identify weak spots, and work towards mastering their technique on these complex movements.

Clean bar path

The clean is a more complex movement, and the bar path will resemble an 'S' shape. The bar starts on the ground, and as you lift, it should stay close to your body. As you reach your hips, you'll extend explosively, causing the bar to come up and slightly away from your body. As you drop under the bar to catch it on your shoulders, the bar will move back towards your body, completing the 'S' shape.

A power clean with bar path. This rep is probably too much forwards and back as you see a forward curve in the path from hip height.

A power clean with a bar path that shows a forward curve from hip height may indicate that the lifter is pulling the bar too far forward, which can lead to missed lift, inefficiency and lost weight on the bar.

Snatch bar path

The snatch bar path is similar to the clean but with a more exaggerated 'S' shape.

The bar starts on the ground and stays close to your body as you lift. As the barbell reaches your hips, you'll extend explosively, ideally the barbell stays as close to vertical as possible, avoiding the hip drive to cause the bar to move forwards and away from your body, leaking energy.

As you drop under the bar to catch it overhead, the bar will move back towards your body, creating a more pronounced 'S' shape.

How can I improve my bar path?

Your bar path is more than just a result of your biomechanics and gravity; it is a detailed and insightful roadmap of your technique.

The importance of a consistent, efficient bar path cannot be overstated. It allows you to maximise the force and power you are able to generate throughout the repetition, ensuring every ounce of energy is used efficiently and goes in the right direction.

Minimising the amount of side-to-side or rotational motion of the bar while also controlling forward and backward movement improves efficiency. With improved efficiency, you can increase the load on the barbell, while also reducing the likelihood of missed lifts, injuries, and wasted training effort.

Use bar path analysis to teach technique basics

Most simply, bar path tracking is an outstanding corrective and teaching tool. The visual representation of bar path helps pinpoint specific areas in lifts where form may be compromised.

Providing the immediate visual feedback for the lifter — especially combined with velocity data — is a highly effective way to instigate real change because bar path taps into the visual learning skills of the lifter.

Are you leading too much with your hips in your deadlift causing the bar to swing away from you? Does the barbell drift forward during your

squat ascent?

With bar path you can confirm these technical flaws, allowing you to make corrective adjustments to your form, or as a coach deliver the right cues and instructions to make the next set sharper.

Use bar path to accelerate advanced training

Consider bar path and video analysis as an assistant coach to fine-tune lifting technique. It is unlikely you will benefit from putting the microscope on every path for every lift, but choosing to dedicate a training block to checking and improving path will pay off.

For example, a powerlifting coach working with a lifter preparing for a competition may use bar path tracking to identify subtle shifts or imbalances during the bench press. This nuance may reveal underlying weaknesses or imbalances in the lifter's core or benching arc. This presents an amazing opportunity to make targeted training interventions which actually matter.

Similarly, for an athlete working on their clean or snatch progressions, tracking bar path can provide concrete feedback on the good and bad repetitions across a set beyond “feel”. This accelerates the critical association between feel and creating good shapes under the bar.

Especially for competitive athletes, there are limits on how much time they have in the weight room. Using bar path tracking is like having an extra set of expert eyes, enabling real-time feedback and immediate correction, enhancing the overall training experience and maximising progress.

Use bar path across a set to gauge consistency

Over the course of a set the bar path tracking in Metric displays a solid green line for the current repetition, with all previous repetitions shown as a grey, faded line.

Comparing each rep across the set helps you more objectively identify inconsistency in your lifts, while simultaneously connecting the change in path to bar speed.

Metric also allows you to save videos to your camera roll where you can then then perform comparisons between sets of the same exercise and across different loads to see how your technique holds up under more challenging loads.

When you should not use bar path tracking

It is important to recognise that bar path tracking may not be appropriate in all training situations.

With novice lifters or during exercises where technique is less critical, an overemphasis on bar path can lead to analysis paralysis or simply consume a lot of valuable training time.

Too much focus on tracking the precise movement of the bar might detract from other essential aspects of training such as feel, intuition, and overall body awareness. Moreover, not every deviation from an "ideal" path indicates a problem, and natural variations in lifting styles could be unnecessarily emphasised.

The key lies in understanding when and how to integrate bar path analysis into your training for optimal results.

Start tracking your bar path with the Metric VBT app today

Bar path is a **free feature** available in the Metric VBT app.

To enable this the new bar path feature, download and open the app, login or create your free profile and open the Account tab then → Settings → Video customisation.

Toggle the bar path switch and voilà – bar path will be applied to all of your lifts allowing you to dive deep into the finer points of your lifting technique!

[Download Metric VBT for iOS here →](#)

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By lifters for lifters

Metric was built with lifters and coaches front of mind.

Focused analytics and powerful workout tracking tools for Powerlifting, Strength and conditioning, CrossFit, Weightlifting, and anyone serious about their strength training.





METRIC

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